UI/UX DESIGN SPECIFICATION

AI-Powered Analytics Intelligence Platform

Wireframes + Full Design Specification

Professional & Corporate | Light + Dark Mode | Tailwind CSS v3

VERSION 1.0 — FINAL

Field	Value
Document	UI/UX Design Specification — Version 1.0 Final
Project	AI-Powered Analytics Intelligence Platform
Stack	React 18 + Vite + Tailwind CSS v3
Style	Professional & Corporate — Blues and Grays
Color Modes	Light Mode + Dark Mode (Tailwind dark: prefix)
Screens Covered	6 screens — Auth, Connect, Dashboard, AI Chat, Admin, Settings
Deliverables	Wireframes + Color Palette + Typography + Components + Screen Specs
Date	March 2, 2026

1. Design Principles

These principles govern every design decision across all screens. Every component, layout, and interaction must be evaluated against them before implementation.

Principle	Description	Applied As
Clarity First	Data must be immediately understandable. No decorative complexity.	Clean cards, generous whitespace, clear hierarchy
Data Density	Show meaningful data without overwhelming. Progressive disclosure for detail.	Summary cards on top, detail tables below, charts in between
Professional Trust	The platform handles business data. It must feel reliable and polished.	Consistent blues, sharp corners on cards, no playful elements
Speed Perception	Every data load must feel fast. Never show a blank area.	Skeleton loaders on every async section before data arrives
AI as Assistant	AI panel supports, not dominates. Data is primary. AI is the helper.	AI panel is right sidebar — never fullscreen by default
Responsive First	Dashboard must work on 375px mobile up to 1920px desktop.	Tailwind responsive prefixes: sm: md: lg: xl: throughout
Accessible	Minimum WCAG AA contrast on all text. Keyboard navigable. Screen reader ready.	4.5:1 text contrast ratio. Focus rings on all interactive elements
Dark Mode Parity	Dark mode is a first-class experience — not an afterthought.	Every component defined in both light and dark variants

2. Color Palette

All colors are defined as Tailwind CSS custom colors in tailwind.config.js. Use ONLY these values throughout the entire application. Never use arbitrary hex values in component files.

2.1 Primary Brand Blues

> 📌 These are the primary interactive and brand colors. Used for buttons, links, active states, borders, and key UI elements.

brand-900 — Deepest Navy	#0D2137	Page backgrounds in dark mode
brand-800 — Dark Navy	#1F4E79	Primary headings, table headers, nav background
brand-700 — Mid Navy	#1A5C9A	Hover states on dark backgrounds
brand-600 — Primary Blue	#2E75B6	Primary buttons, active nav, links, chart primary line
brand-500 — Medium Blue	#3B8ED4	Button hover, secondary chart lines
brand-400 — Light Blue	#6EB2E8	Focus rings, badge backgrounds, light mode accents
brand-100 — Pale Blue	#D6E8F8	Card backgrounds (light), table row alt, tag backgrounds
brand-50 — Near White Blue	#EEF6FD	Subtle section backgrounds, hover states on white

2.2 Neutral Grays

> 📌 Neutrals form the backbone of the layout. All text, borders, and backgrounds use these values.

neutral-900 — Near Black	#111827	Primary body text (light mode), page bg (dark mode alt)
neutral-800 — Dark Gray	#1F2937	Secondary headings, sidebar bg (dark mode)
neutral-700 — Medium Dark	#374151	Body text secondary, icon fill
neutral-600 — Medium Gray	#4B5563	Placeholder text, disabled labels
neutral-400 — Light Gray	#9CA3AF	Border colors, dividers, input borders
neutral-200 — Very Light	#E5E7EB	Card borders (light mode), table dividers
neutral-100 — Off White	#F3F4F6	Page background (light mode), sidebar (light)
neutral-50 — Near White	#F9FAFB	Card background (light mode)

2.3 Semantic Colors

semantic-success — Green	#16A34A	Positive % change, success toasts, connected badges
semantic-warning — Amber	#D97706	Warning toasts, expiring token alerts, caution badges
semantic-error — Red	#DC2626	Negative % change, error toasts, disconnected badges, validation errors
semantic-info — Blue	#2E75B6	Informational toasts, help tips

2.4 Dark Mode Surface Colors

dark-bg — Deepest Dark #0F172A | class: dark:bg-dark-bg — main page background

dark-surface — Dark Surface #1E293B | class: dark:bg-dark-surface — sidebar, nav

dark-card — Dark Card #273549 | class: dark:bg-dark-card — card backgrounds

2.5 Tailwind Config — Color Tokens

Add these exact values to tailwind.config.js theme.extend.colors:

```
brand: {
  50: '#EEF6FD',  100: '#D6E8F8',  400: '#6EB2E8',
  500: '#3B8ED4', 600: '#2E75B6',  700: '#1A5C9A',
  800: '#1F4E79', 900: '#0D2137'
},
neutral: {
  50: '#F9FAFB',  100: '#F3F4F6',  200: '#E5E7EB',
  400: '#9CA3AF', 600: '#4B5563',  700: '#374151',
  800: '#1F2937', 900: '#111827'
},
semantic: { success:'#16A34A', warning:'#D97706', error:'#DC2626', info:'#2E75B6' },
dark: { bg:'#0F172A', surface:'#1E293B', card:'#273549' }
```

3. Typography

The platform uses Inter as the primary font. It is clean, highly legible at small sizes, and widely used in enterprise SaaS products.

3.1 Font Setup

- Font family: Inter (Google Fonts)
- Import in index.html: <link href='https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap' rel='stylesheet'>
- Set in tailwind.config.js: fontFamily: { sans: ['Inter', 'ui-sans-serif', 'system-ui'] }
- Monospace (for code/API keys): font-mono — Tailwind default

3.2 Type Scale

Token	Size	Weight	Line Height	Tailwind Class	Used For
Display	36px	700 Bold	1.2	text-4xl font-bold	Page titles on auth screens
H1	30px	700 Bold	1.3	text-3xl font-bold	Section headings
H2	24px	600 SemiBold	1.4	text-2xl font-semibold	Card headings, tab labels
H3	20px	600 SemiBold	1.4	text-xl font-semibold	Sub-section headings
H4	16px	600 SemiBold	1.5	text-base font-semibold	Card sub-titles, table col headers
Body-LG	16px	400 Regular	1.6	text-base font-normal	Main body text, descriptions
Body	14px	400 Regular	1.6	text-sm font-normal	Table rows, labels, secondary text
Body-SM	12px	400 Regular	1.5	text-xs font-normal	Captions, helper text, timestamps
KPI Value	32px	700 Bold	1.1	text-3xl font-bold	KPI card main metric values
KPI Change	13px	500 Medium	1.4	text-xs font-medium	% change badges on KPI cards
Badge	11px	600 SemiBold	1.3	text-xs font-semibold	Status badges, tags
Code	13px	400 Regular	1.6	text-sm font-mono	API keys in admin, code snippets

4. Component Library

All reusable components are defined here. Every component must be implemented in both light and dark mode. Components live in /client/src/components/ui/.

4.1 Buttons

Variant	Light Mode Classes	Dark Mode Classes	Use Case
Primary	bg-brand-600 text-white hover:bg-brand-700 focus:ring-2 focus:ring-brand-400	dark:bg-brand-500 dark:hover:bg-brand-600	Main actions: Save, Connect, Submit
Secondary	bg-white text-brand-700 border border-brand-300 hover:bg-brand-50	dark:bg-dark-card dark:text-brand-400 dark:border-brand-700	Secondary actions: Cancel, Back
Danger	bg-semantic-error text-white hover:bg-red-700 focus:ring-2 focus:ring-red-400	dark:bg-red-700 dark:hover:bg-red-800	Destructive: Delete, Disconnect
Ghost	bg-transparent text-brand-600 hover:bg-brand-50 focus:ring-2 focus:ring-brand-200	dark:text-brand-400 dark:hover:bg-dark-card	Tertiary: Learn more, view all
Icon-only	p-2 rounded-lg text-neutral-600 hover:bg-neutral-100 focus:ring-2 focus:ring-brand-400	dark:text-neutral-400 dark:hover:bg-dark-card	Icon buttons in headers

> 📌 All buttons: rounded-lg px-4 py-2 text-sm font-medium transition-colors duration-150 focus:outline-none focus:ring-offset-2. Disabled state: opacity-50 cursor-not-allowed pointer-events-none.

4.2 Input Fields

State	Light Mode Classes	Dark Mode Classes
Default	w-full px-3 py-2 border border-neutral-300 rounded-lg text-sm bg-white text-neutral-900 focus:outline-none focus:ring-2 focus:ring-brand-400 focus:border-brand-400	dark:bg-dark-card dark:border-neutral-600 dark:text-white dark:placeholder-neutral-500
Error	border-semantic-error focus:ring-semantic-error	dark:border-red-500
Disabled	bg-neutral-100 text-neutral-400 cursor-not-allowed opacity-60	dark:bg-dark-surface dark:text-neutral-600
Password	Same as Default + pr-10 for show/hide toggle button	Same dark mode as Default

4.3 KPI Card

Element	Light Mode	Dark Mode	Notes
Card container	bg-white rounded-xl border border-neutral-200 p-5 shadow-sm	dark:bg-dark-card dark:border-neutral-700	shadow-sm — subtle elevation
Icon container	w-10 h-10 rounded-lg bg-brand-100 flex items-center justify-center	dark:bg-brand-900	Icon color: text-brand-600
Metric value	text-3xl font-bold text-neutral-900 mt-3	dark:text-white	Use tabular-nums class for number alignment
Metric label	text-sm font-medium text-neutral-500 mt-1	dark:text-neutral-400	e.g. 'Total Users'
Positive change	inline-flex items-center text-xs font-medium text-semantic-success bg-green-50 px-2 py-0.5 rounded-full	dark:bg-green-900 dark:text-green-400	+12.4% with arrow-up icon
Negative change	inline-flex items-center text-xs font-medium text-semantic-error bg-red-50 px-2 py-0.5 rounded-full	dark:bg-red-900 dark:text-red-400	-3.2% with arrow-down icon
Skeleton loader	animate-pulse bg-neutral-200 rounded h-8 w-24	dark:bg-neutral-700	Show while data is loading

4.4 Data Table

Element	Classes	Dark Mode
Table wrapper	w-full overflow-x-auto rounded-xl border border-neutral-200	dark:border-neutral-700
Table	w-full text-sm text-left border-collapse	
Header row	bg-neutral-100 text-neutral-600 text-xs font-semibold uppercase tracking-wider	dark:bg-dark-surface dark:text-neutral-400
Header cell	px-4 py-3 border-b border-neutral-200 cursor-pointer hover:bg-neutral-200 select-none	dark:border-neutral-700 dark:hover:bg-neutral-700
Sort indicator	ml-1 text-brand-600 inline	dark:text-brand-400
Body row even	bg-white hover:bg-brand-50 transition-colors	dark:bg-dark-card dark:hover:bg-dark-surface
Body row odd	bg-neutral-50 hover:bg-brand-50 transition-colors	dark:bg-dark-surface dark:hover:bg-neutral-700
Body cell	px-4 py-3 border-b border-neutral-100 text-neutral-800	dark:border-neutral-700 dark:text-neutral-200

4.5 Badge / Status Pill

Variant	Classes	Use Case
Connected / Success	bg-green-100 text-green-800 text-xs font-semibold px-2.5 py-0.5 rounded-full dark:bg-green-900 dark:text-green-300	Source connected, email verified
Disconnected / Error	bg-red-100 text-red-800 text-xs font-semibold px-2.5 py-0.5 rounded-full dark:bg-red-900 dark:text-red-300	Source not connected, error state
Pending / Warning	bg-amber-100 text-amber-800 text-xs font-semibold px-2.5 py-0.5 rounded-full dark:bg-amber-900 dark:text-amber-300	Token expiring soon, pending
Neutral / Info	bg-brand-100 text-brand-800 text-xs font-semibold px-2.5 py-0.5 rounded-full dark:bg-brand-900 dark:text-brand-300	Informational labels, source type

4.6 Toast Notifications

Type	Icon	Background	Position	Duration
Success	CheckCircle	bg-green-600 text-white	Top-right, fixed	3 seconds auto-dismiss
Error	XCircle	bg-red-600 text-white	Top-right, fixed	6 seconds — user must dismiss or wait
Warning	AlertTriangle	bg-amber-500 text-white	Top-right, fixed	5 seconds auto-dismiss
Info	Info	bg-brand-600 text-white	Top-right, fixed	3 seconds auto-dismiss

> 📌 Use react-hot-toast library. Configure in App.jsx: <Toaster position='top-right' toastOptions={{ duration: 3000 }} />

4.7 Skeleton Loader

- Every card, chart, and table must show a skeleton loader while data is fetching
- Skeleton: animate-pulse bg-neutral-200 rounded dark:bg-neutral-700
- KPI card skeleton: 3 gray bars at heights h-4, h-8, h-4 with spacing
- Table skeleton: 5 rows of h-10 bars with full width
- Chart skeleton: single h-48 bar filling the chart container
- Never show empty white space — skeleton must appear within 50ms of component mount

5. Spacing System & Layout Grid

5.1 Spacing Scale

Use only Tailwind's default spacing scale. Never use arbitrary values like mt-[13px].

Token	Size	Use Case
space-1 / 4px	4px	Icon inner padding, tight element gaps
space-2 / 8px	8px	Badge padding, compact list item gaps
space-3 / 12px	12px	Input padding (py), small card inner spacing
space-4 / 16px	16px	Standard gap between elements, card padding
space-5 / 20px	20px	Card padding (p-5), section gaps
space-6 / 24px	24px	Gap between cards in a row, section padding
space-8 / 32px	32px	Section top/bottom padding, large gaps
space-10 / 40px	40px	Page section separators
space-12 / 48px	48px	Major layout section spacing
space-16 / 64px	64px	Page-level top padding, hero sections

5.2 Layout Grid

Breakpoint	Min Width	Columns	Gutter	Max Content Width
Mobile (default)	0px	1 col	16px	100%
sm	640px	2 cols	16px	640px
md	768px	2 cols	24px	768px
lg	1024px	12 cols	24px	1024px
xl	1280px	12 cols	24px	1280px
2xl	1536px	12 cols	32px	1400px (max-w-7xl)

5.3 Dashboard Layout Columns

Viewport	Left Column (Analytics)	Right Column (AI Panel)	Nav
Mobile (<768px)	Full width — 100%	Hidden — accessible via tab/button	Bottom nav bar
Tablet (768px)	60% width	40% width — collapsible	Left sidebar, collapsed to icons
Desktop (1024px)	65% width	35% width	Left sidebar, expanded with labels

Viewport	Left Column (Analytics)	Right Column (AI Panel)	Nav
Wide (1280px+)	70% width	30% width (min 320px)	Left sidebar, expanded

6. Wireframes — All Screens

Wireframes below use ASCII box notation to show layout structure. These are low-fidelity — they define structure, hierarchy, and component placement, not visual styling. Numbers in brackets [1] refer to design notes below each wireframe.

6.1 Login Page

📌 Route: /login | Unauthenticated only. If user has valid JWT → redirect to /dashboard

```
┌──────────── FULL PAGE — centered vertically + horizontally ────────────┐
│  [LOGO]  AI Analytics Platform                  [Light/Dark Toggle]    │
├────────────────────────────────────────────────────────────────────────┤
│                        CARD (max-w-md, centered)                       │
│  ┌──────────────────────────────────────────────────────────────────┐  │
│  │  Welcome back                                        [1]         │  │
│  │  Sign in to your account                                         │  │
│  │                                                                  │  │
│  │  [ Email address input                                ]          │  │
│  │  [ Password input                          [show]   ]  [2]       │  │
│  │                                Forgot password?   [3]            │  │
│  │                                                                  │  │
│  │  [ ═══════  Sign In  ═══════ ]  (primary button, full width)     │  │
│  │                                                                  │  │
│  │  ──────────────────  or  ──────────────────────────              │  │
│  │                                                                  │  │
│  │  [ G  Continue with Google  ]  (full width, outlined)  [4]       │  │
│  │                                                                  │  │
│  │  Don't have an account?  Sign up →              [5]              │  │
│  └──────────────────────────────────────────────────────────────────┘  │
└────────────────────────────────────────────────────────────────────────┘
```

Ref	Element	Behaviour / Detail
[1]	Page heading	text-3xl font-bold text-neutral-900. Subtitle: text-sm text-neutral-500
[2]	Password field	type=password with eye icon toggle button (absolute right-3). Uses @heroicons/react EyeIcon / EyeSlashIcon
[3]	Forgot password	text-sm text-brand-600 hover:text-brand-800 float-right above Sign In button
[4]	Google button	White bg, border-neutral-300, Google colored G icon SVG. Calls GET /api/auth/google
[5]	Sign up link	Inline link — navigates to /register via React Router Link
BG	Page background	Light: bg-neutral-100. Dark: bg-dark-bg. Card: bg-white dark:bg-dark-card shadow-lg rounded-2xl p-8

6.2 Register Page

📌 Route: /register | Same card layout as Login. On success → show 'Check your email' message

```
| CARD (same container as login)                                  |
| ┌──────────────────────────────────────────────────────────┐   |
| | Create your account                                      |   |
| |                                                          |   |
| | [ Full name input                                ]       |   |
| | [ Email address input                            ]       |   |
| | [ Password input                        [show]   ]       |   |
| | [ Confirm password                      [show]   ]       |   |
| |                                                          |   |
| | Password strength indicator: ▓▓▓
```

```
│  └──────────────
```

```
│  └──────────────────────────┘  └──────────────────────────┘            │
│                                                                         │
│  [ Skip for now ]           [ Continue to Dashboard → ]                 │
└─────────────────────────────────────────────────────────────────────────┘
```

Element	Detail
Connection cards	2-column grid on desktop, stack on mobile. Each card: rounded-2xl border-2. Default: border-neutral-200. Connected: border-semantic-success bg-green-50
Status dots	Green filled circle = connected. Gray empty circle = not connected. Show which sub-services are active (GA4, GSC, Ads separately for Google card)
Connect button	Primary button when not connected. On connected: shows green 'Connected ✓' badge + 'Disconnect' ghost button
After Google OAuth	Page refreshes OAuth status — shows which of GA4, GSC, Google Ads are available from that account
Continue button	Disabled (opacity-50) if no sources connected. Enabled once at least 1 connected
Skip link	Ghost text button — bottom left. Goes directly to dashboard with empty state

6.4 Main Dashboard

> 📌 Route: /dashboard | Protected. This is the primary screen. Contains left sidebar nav, main content area with tabs, and right AI panel.

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│ LEFT SIDEBAR (w-64)  │  TOP BAR                                                    │  [User Avatar ▼] │
│                      │  [Date Range Picker] [Refresh]                              │  [Dark Mode ☾]   │
├──────────────────────┼──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┤
│  [Logo]              │  SOURCE TABS: Overview|GA4|GSC|Google Ads|Facebook          │
│                      │                                                             │
│  ● Dashboard         │  KPI CARDS ROW (4 cards per source)                         │
│                      │                                                             │
│  Connect Accounts    │  ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────┐
```

```
│   Settings            │  │ Users    │ │Sessions │ │Bounce % │ │Avg Time │
                                      │
│                       │  │ 12,450   │ │ 18,230  │ │  42.3%  │ │  2m 34s │
                                      │
│ ──────────────        │  │ +12.4%↑  │ │  +8.1%↑ │ │  -2.1%↓ │ │         │
                                      │
│ [Admin] (super admin)│  └──────────┘ └─────────┘ └─────────┘ └─────────┘
                                      │
│                       │──────────────────────────────────────── AI PANEL
                                   ──│
│                       │  CHART (Line/Bar — 65% width)     │ Weekly Insight
                                      │
     │                     │  ──────────────────────────     │ card
                                      │
        │                    │                                 │
                             ──────────────── │
│                       │  TOP TABLE (sortable, 10 rows)    │ AI Chat Panel
                                      │
│                       │  Col1  Col2  Col3  Col4  Col5     │ [chat history]
                                      │
        │                    │  ....  ....  ....  ....  ....     │
                                      │
     │                     │                                 │ [Ask
                                anything…] │
│                       │  SECOND CHART (if applicable)     │ [Suggestions]
                                      │
└──────────────────────┴──────────────────────────────────┴──────────────
                                   ──┘
```

Element	Specification
Left Sidebar	w-64 fixed height. bg-brand-800 text-white. Active item: bg-brand-700 rounded-lg. Hover: bg-brand-700/50. Admin link only visible to super admin email.
Top Bar	h-16 bg-white dark:bg-dark-surface border-b border-neutral-200 dark:border-neutral-700. Contains date range picker and user menu.
Date Range Picker	Presets: 7d, 28d, 30d, 90d, Custom. Dropdown style. On change: all dashboard data refetches. Persist selection in Zustand + localStorage.
Source Tabs	Tab bar below top bar. Active tab: border-b-2 border-brand-600 text-brand-600. Each tab shows connection status dot. Disconnected tab: grayed out with 'Connect' chip.

Element	Specification
KPI Cards Row	4-column grid: grid-cols-2 md:grid-cols-4 gap-4. Responsive — 2 cols on tablet, 4 on desktop.
Main Chart	Line chart for timeseries. Recharts ResponsiveContainer width=100% height=280. Tooltip on hover. Legend below chart.
Data Tables	Full-width below chart. Sortable columns. Pagination: show 10 rows, 'Load more' button. Sticky header on scroll.
AI Panel	Right side w-80 xl:w-96 fixed height with internal scroll. bg-white dark:bg-dark-card border-l border-neutral-200 dark:border-neutral-700.

6.4.1 Overview Tab — KPI Cards

Card	Metric	Icon	Change Logic
GA4 — Users	activeUsers	UserGroupIcon	Green if positive
GA4 — Sessions	sessions	CursorArrowRaysIcon	Green if positive
GSC — Clicks	clicks	CursorClickIcon	Green if positive
Google Ads — Spend	cost (formatted)	BanknotesIcon	Neutral — context dependent
Google Ads — ROAS	roas	ChartBarIcon	Green if > 1, red if < 1
FB — Reach	reach	MegaphoneIcon	Green if positive
FB — Spend	spend (formatted)	BanknotesIcon	Neutral
FB — ROAS	roas	ChartBarIcon	Green if > 1, red if < 1

6.5 AI Chat Panel (Right Panel)

> 📌 The AI panel is always visible on desktop as the right column. On mobile it slides up as a sheet from the bottom.

```
┌── AI PANEL (w-80 fixed right) ──────────────────────────────┐
│  PANEL HEADER                                                 │
│  [✦ AI Assistant]                [New Chat] [History ▼]  [×]  │
├──────────────────────────────────────────────────────────────┤
│  WEEKLY INSIGHT CARD                                          │
│  ┌──────────────────────────────────────────────────────┐    │
│  │  ✦ Weekly Summary  — Generated today at 9:04 AM  [Refresh ↺]  │  │
│  │  Your traffic grew 12% this week driven by organic search...  │  │
│  │  Google Ads ROAS improved to 3.4x...                          │  │
```

```
│  └──────────────────────────────────────────────────────────────────────┘  │
├──────────────────────────────────────────────────────────────────────────────┤
│  CHAT HISTORY (scrollable)                                                   │
│                                                                              │
│                                          ┌──────────────────────────┐        │
│                  [User Avatar]           │ Why did traffic drop?    │        │
│                                          └──────────────────────────┘        │
│  ┌──────────────────────────────────────────────┐                           │
│  │ [✦]  Traffic dropped 18% on Jan 15th due     │  [AI Avatar]              │
│  │       to a Google algorithm update...        │                           │
│  │       **Recommendation:** Review your        │                           │
│  │       top landing pages for thin content.    │                           │
│  └──────────────────────────────────────────────┘                           │
├──────────────────────────────────────────────────────────────────────────────┤
│  SUGGESTED QUESTIONS                                                         │
│  [Why did bounce rate increase?]  [Which ads have best ROAS?]                │
│  [What is our top search query?]  [Compare this week vs last]                │
├──────────────────────────────────────────────────────────────────────────────┤
│  INPUT AREA                                                                  │
│  ┌──────────────────────────────────────────────────────────────────┐        │
│  │ Ask anything about your analytics...                             │        │
│  │                                                                  │        │
│  └──────────────────────────────────────────────────  [Send ▶]  ──┘         │
│  Ctrl+Enter to send                                                          │
└──────────────────────────────────────────────────────────────────────────────┘
```

Element	Specification
Panel header	h-12 border-b. Title: '✦ AI Assistant' in text-brand-600 font-semibold. Buttons: New Chat (ghost), History dropdown, collapse X
Weekly insight card	bg-brand-50 dark:bg-brand-900/30 border border-brand-200 dark:border-brand-

Element	Specification
	700 rounded-xl p-4. Prose text: text-sm. Max 5 lines shown, 'Read more' expands.
User message	Right-aligned. bg-brand-600 text-white rounded-2xl rounded-tr-none px-4 py-2.5 max-w-xs text-sm
AI message	Left-aligned. bg-white dark:bg-dark-surface border border-neutral-200 dark:border-neutral-700 rounded-2xl rounded-tl-none px-4 py-3 max-w-xs. Markdown rendered.
AI loading state	Animated 3-dot pulse bubble in AI message position: '● ● ●' with fade animation
Suggestions	Horizontal scroll row of pill chips: bg-neutral-100 dark:bg-dark-card border rounded-full px-3 py-1.5 text-xs font-medium. Click → prefill input
Input textarea	resize-none rows=2. Expands to rows=4 max on typing. Send button right-aligned inside input container. Disabled + spinner while AI is responding.
Markdown rendering	react-markdown + remark-gfm. Styles: headings text-sm font-semibold, bold font-semibold, bullets list-disc list-inside, code font-mono bg-neutral-100 rounded px-1

6.6 Super Admin Panel

> 📌 Route: /dashboard/admin | Admin tab in left sidebar. Visible ONLY to super admin email. Same dashboard shell — replaces main content area.

```
┌──────────────────────────────────────────────────────────────────────┐
│  PAGE HEADING: Platform API Configuration                            │
│  SUBHEADING: Manage API keys for all integrations from this screen   │
│                                                                      │
│  ⚠️ Warning banner: Changes take effect immediately for all users    │
│                                                                      │
│  ┌── GOOGLE INTEGRATION ──────────────────── [● 2/3 Configured] ──┐  │
│  │  [▼ Collapse]                                                  │  │
│  │                                                                │  │
│  │  Google OAuth Client ID                  [● Configured]        │  │
│  │  [ 123456****7890.apps.google... ]  [Show] [Edit]  [Test ✓]    │  │
│  │  Last updated: Jan 12 2025 by admin@company.com                │  │
│  │                                                                │  │
│  │  Google OAuth Client Secret              [● Configured]        │  │
│  │  [ GOCS****              ]  [Show] [Edit]  [Test ✓]            │  │
```

```
│  │                                                          │ │
│  │  Google Ads Developer Token            [○ Not Set]       │ │
│  │  [ Enter Google Ads developer token...  ]  [Save]  [Test] │ │
│  │                                                          │ │
│  │                        [ Save All Google Settings ]      │ │
│  └──────────────────────────────────────────────────────────┘ │
│                                                               │
│  ┌── FACEBOOK INTEGRATION ──────────────── [● 2/2 Configured] ──┐ │
│  │  Facebook App ID + Facebook App Secret  (same row pattern)   │ │
│  └──────────────────────────────────────────────────────────┘ │
│                                                               │
│  ┌── AI SERVICES ──────────────────────── [● 1/2 Configured] ──┐ │
│  │  Anthropic Claude API Key + OpenAI API Key (fallback)        │ │
│  └──────────────────────────────────────────────────────────┘ │
│                                                               │
│  ┌── OTHER SERVICES ──────────────────── [● 1/1 Configured] ──┐ │
│  │  Resend Email API Key                                        │ │
│  └──────────────────────────────────────────────────────────┘ │
└──────────────────────────────────────────────────────────────┘
```

Element	Specification
Page warning	amber bg-amber-50 border-l-4 border-amber-500 rounded-lg p-4 mb-6. Icon: AlertTriangle. Text in text-amber-800.
Group sections	Each group is a collapsible card. Header: flex justify-between items-center. Collapse icon rotates 180deg on open. Default: all expanded.
Group badge	'X/Y Configured' pill — green if all configured, amber if partial, red if none. Shows count of configured vs total keys in group.
Key row	3-column layout: label + status badge | masked value input | action buttons
Masked input	font-mono text-sm bg-neutral-100 dark:bg-dark-surface rounded-lg px-3 py-2. Read-only. 'Edit' button switches to editable password input.
Edit mode	Clicking Edit: input becomes type=password editable. Show/hide toggle appears. Cancel button restores masked view. Save button calls API.

Element	Specification
Test button	Ghost button with plug icon. On click: shows spinner → then green '✓ Connected' or red '✗ Failed: {error}'
Save All button	Primary button at bottom of each group. Collects all edited values in group, calls POST /api/admin/config/bulk
Not configured	Gray badge. Input shows placeholder text. Save button prominent.

6.7 Settings Page

📌 Route: /dashboard/settings | Accessible from left nav and user avatar menu.

```
┌──────────────────────────────────────────────────────────────────────────┐
│  Settings                                                                │
│                                                                          │
│  ┌── PROFILE ──────────────────────────────────────────────────────────┐ │
│  │  [Avatar]  John Doe                                                 │ │
│  │            john@example.com  [● Email Verified]                     │ │
│  │            [ Display Name input          ]  [ Save Changes ]        │ │
│  └─────────────────────────────────────────────────────────────────────┘ │
│                                                                          │
│  ┌── CONNECTED ACCOUNTS ───────────────────────────────────────────────┐ │
│  │  [G]  Google Account (GA4 + GSC + Ads)  [● Connected]  [Disconnect]│
│  │        Connected as: john@gmail.com                                 │ │
│  │  [f]  Facebook Account (Meta Ads)       [● Connected]  [Disconnect]│
│  │        Connected as: John Doe (Facebook) | Token expires: 45 days| │
│  └─────────────────────────────────────────────────────────────────────┘ │
│                                                                          │
│  ┌── APPEARANCE ───────────────────────────────────────────────────────┐ │
│  │  Theme:  [● Light]  [  Dark]  [  System]                            │ │
│  └─────────────────────────────────────────────────────────────────────┘ │
│                                                                          │
│  ┌── DANGER ZONE ──────────────────────────────────────────────────────┐ │
```

```
│  │  bg-red-50 border border-red-200                                  │ │
│  │  Delete Account                                                   │ │
│  │  This permanently deletes your account and ALL data.              │ │
│  │                                [ Delete My Account ] (danger btn)  │ │
│  └──────────────────────────────────────────────────────────────────┘ │
└────────────────────────────────────────────────────────────────────────┘
```

Element	Specification
Settings sections	Each section is a white card rounded-xl border p-6 dark:bg-dark-card. Stacked vertically with gap-6.
Avatar	w-16 h-16 rounded-full bg-brand-600 text-white text-xl font-bold flex items-center justify-center. Shows initials if no Google photo.
Disconnect button	Danger variant button. Shows confirmation modal: 'Disconnect Google? This will remove GA4, GSC, and Google Ads data.' with Cancel + Confirm buttons.
Facebook token	If token expires < 14 days: show amber warning: 'Token expiring in X days. Reconnect now.' with Reconnect button.
Theme toggle	3-way radio button group styled as toggle pills. Saves preference to localStorage + Zustand. System option reads prefers-color-scheme.
Delete account	Danger zone card: bg-red-50 border-red-200. Delete button opens confirmation modal requiring user to type their email to confirm.
Delete modal	Modal: 'Type your email address to confirm deletion: [input]'. Submit disabled until input matches user email exactly.

7. Interaction Patterns & Micro-Interactions

7.1 Loading States

Scenario	Pattern	Implementation
Initial page load	Full skeleton layout — nav + cards + charts all skeletonized	Show skeleton immediately, replace with data on resolve
Tab switch	Skeleton for content area only — nav and header stay visible	Zustand loading flag per source tab
Date range change	Overlay shimmer on existing data — don't replace with skeleton	opacity-50 pointer-events-none on card containers + spinner in top bar
AI thinking	Animated 3-dot typing indicator in AI message position	CSS keyframe animation on 3 dots with staggered delays
Button submit	Replace button text with spinner icon — button stays same size	disabled + spinner while request is pending
Chart data loading	Gray placeholder rectangle same height as chart	animate-pulse h-64 bg-neutral-200 dark:bg-neutral-700 rounded-xl

7.2 Empty States

Scenario	Display
No data for date range	Centered illustration (simple SVG) + 'No data for this period' + 'Try a wider date range' suggestion
Source not connected	Card with source logo, short description, and 'Connect [Source]' primary button — no error language
No AI conversations	Center of chat panel: icon + 'Ask your first question' + 3 suggested question chips
No search results	In tables: 'No results found' with clear filters suggestion

7.3 Transitions & Animations

Element	Animation	Tailwind / CSS
Page transitions	Fade in on route change	opacity-0 → opacity-100, duration-200
Modal open/close	Scale + fade	scale-95 opacity-0 → scale-100 opacity-100, duration-150
Sidebar collapse	Slide left with width transition	transition-all duration-300 ease-in-out
Toast appear	Slide in from right	translate-x-full → translate-x-0, duration-300

Element	Animation	Tailwind / CSS
Tab switch	Underline slides to active tab	transition-transform duration-200 on indicator bar
Accordion expand	Height animates open	max-h-0 → max-h-screen, overflow-hidden, duration-300
Button hover	Background color shift	transition-colors duration-150
KPI card hover	Subtle lift with shadow	hover:shadow-md transition-shadow duration-150

> 📌 Keep all animations under 300ms. Never animate purely decorative elements. Respect prefers-reduced-motion: @media (prefers-reduced-motion) { * { transition: none !important; animation: none !important; } }

8. Dark Mode Specification

Dark mode uses Tailwind's class strategy: add class='dark' to the <html> element when dark mode is active. Controlled via Zustand themeStore + localStorage.

8.1 Dark Mode Color Mapping

Element	Light Mode	Dark Mode
Page background	bg-neutral-100	dark:bg-dark-bg (#0F172A)
Sidebar / Nav	bg-brand-800	dark:bg-dark-surface (#1E293B)
Card background	bg-white	dark:bg-dark-card (#273549)
Top bar	bg-white	dark:bg-dark-surface
Primary text	text-neutral-900	dark:text-white
Secondary text	text-neutral-500	dark:text-neutral-400
Border / Divider	border-neutral-200	dark:border-neutral-700
Input background	bg-white	dark:bg-dark-card
Table header	bg-neutral-100	dark:bg-dark-surface
Table row alt	bg-neutral-50	dark:bg-dark-surface
Skeleton loader	bg-neutral-200	dark:bg-neutral-700
Code / Mono bg	bg-neutral-100	dark:bg-neutral-800
AI panel bg	bg-white	dark:bg-dark-card
Insight card bg	bg-brand-50	dark:bg-brand-900/30

8.2 Dark Mode Implementation

- Set strategy in tailwind.config.js: darkMode: 'class'
- Toggle: document.documentElement.classList.toggle('dark')
- Persist in localStorage key: 'theme' — values: 'light' | 'dark' | 'system'
- System: window.matchMedia('(prefers-color-scheme: dark)').matches
- Apply on app load before render to prevent flash of wrong theme
- All chart colors must also change in dark mode — use Recharts custom colors from Zustand theme

8.3 Chart Colors — Light vs Dark

Chart Element	Light Mode	Dark Mode
Primary line	#2E75B6 (brand-600)	#6EB2E8 (brand-400)
Secondary line	#16A34A (success)	#4ADE80 (green-400)

Chart Element	Light Mode	Dark Mode
Tertiary line	#D97706 (warning)	#FCD34D (amber-300)
Grid lines	#E5E7EB (neutral-200)	#374151 (neutral-700)
Axis labels	#6B7280 (neutral-500)	#9CA3AF (neutral-400)
Tooltip bg	#FFFFFF	#273549 (dark-card)
Tooltip border	#E5E7EB	#4B5563 (neutral-600)
Bar fill	#2E75B6	#3B8ED4
Pie/Donut	brand-600, success, warning, neutral-400	brand-400, green-400, amber-300, neutral-600

9. Responsive Design Rules

Component	Mobile (<768px)	Tablet (768–1023px)	Desktop (1024px+)
Left Sidebar	Hidden — bottom nav bar instead	Collapsed to icons only (w-16)	Expanded with labels (w-64)
KPI Cards	2-column grid	2-column grid	4-column grid
Charts	Full width, h-48	Full width, h-56	65% width, h-64
AI Panel	Hidden — floating button to open as sheet	Collapsible — toggle button in top bar	Always visible right column
Data Tables	Horizontal scroll, sticky first column	Horizontal scroll	Full width, all columns visible
Dashboard tabs	Horizontal scroll row of tabs	All tabs visible	All tabs visible
Admin panel	Stacked single column sections	Stacked single column	Stacked single column (wide cards)
Auth cards	Full width with padding px-4	Max-w-md centered	Max-w-md centered
Settings	Full width stacked sections	Max-w-2xl centered	Max-w-2xl centered

10. Accessibility Requirements

Requirement	Standard	Implementation
Color contrast — body text	WCAG AA: 4.5:1 minimum	neutral-900 on white = 15:1 ✓. neutral-600 on white = 5.9:1 ✓
Color contrast — large text	WCAG AA: 3:1 minimum	All headings pass at defined sizes
Focus indicators	Visible on all interactive elements	focus:ring-2 focus:ring-brand-400 focus:ring-offset-2 on all buttons/inputs
Keyboard navigation	Full keyboard support	Tab order follows visual order. Modal traps focus. Escape closes modal.
Screen reader labels	All inputs labelled	htmlFor + id on all form fields. aria-label on icon buttons.
Error messages	Programmatically associated	aria-describedby linking input to error paragraph
Loading states	Announced to screen readers	aria-live='polite' on data containers. aria-busy='true' while loading.
Chart accessibility	Data tables as fallback	Each chart has an accessible data table hidden visually but available to screen readers
Reduced motion	Respect user preference	@media (prefers-reduced-motion) disables all transitions and animations

— End of Document —

UI/UX Design Specification | Version 1.0 | Final | AI-Powered Analytics Platform

Authored with Claude.ai | Built for Cursor Agentic Development